# OrdinalFed: Federated Learning with Ordinal-Aware Aggregation for Diabetic Retinopathy Grading

***Abstract*—Diabetic Retinopathy (DR) is a leading cause of preventable blindness. With federated learning (FL), hospitals can train one DR grading model together without sharing retinal images. When hospitals see very different patients, however, FedAvg can collapse. This paper studies OrdinalFed, which has two parts. The first is an *OrdinalLoss* that penalises a wrong grade by its squared distance from the true grade. The second is a *QWK-weighted aggregation*: the server scores each client model by its Quadratic Weighted Kappa (QWK) on 50 held-out images, and over the rounds it moves the aggregation weights from image counts towards these scores. All experiments on the DDR dataset were repeated with three seeds and scored on the held-out test split. Under extreme label skew (Dirichlet $\alpha = 0.1$), FedAvg collapsed in every seed, with a worst-round test QWK of 0.378. OrdinalFed raised the worst round to 0.514, with no change at the hospitals and about 1.2 s of extra server time per round. The ablations show where this gain comes from: small hospitals get more weight. Equal client weights did even better (0.630), while the QWK scores and the OrdinalLoss added no measurable benefit. FedProx was the most stable method (0.641), and OrdinalFed with FedProx's proximal term came close to it (0.627). On the IID and realistic multi-site partitions, all methods were within 0.01 QWK of each other. On the external APTOS 2019 dataset, all models scored 0.597–0.658 QWK, close to the centralised model (0.641).**

***Index Terms*—Diabetic Retinopathy, Federated Learning, Non-IID, Ordinal Loss, QWK-Weighted Aggregation, EfficientNet-B3, Medical Imaging**

## I. Introduction

Diabetic Retinopathy (DR) is a common complication of diabetes and one of the main causes of preventable blindness in working-age adults. It damages the small blood vessels of the retina. In fundus photographs, this damage shows up as microaneurysms, haemorrhages and hard exudates, and in advanced cases as the growth of new vessels. Bangladesh alone has more than 13.1 million people with diabetes [1], and outside the big cities there are few retinal specialists. DR is graded on a five-level scale: Grade 0 (no DR), Grade 1 (mild non-proliferative DR, NPDR), Grade 2 (moderate NPDR), Grade 3 (severe NPDR), and Grade 4 (proliferative DR). Referable disease (Grade $\geq 2$) must be found early to prevent vision loss, so automated screening is very useful.

Deep convolutional networks grade DR well when all images can be collected in one place. On the imbalanced DDR dataset [2], an EfficientNet-B3 model trained this way reaches a Quadratic Weighted Kappa (QWK) of 0.834 on the validation split and 0.682 on the separate test split (Sec. V). QWK is the usual metric for ordered grades. It penalises an error by the square of its distance, so calling a Grade 4 image Grade 0 costs much more than calling it Grade 3. In practice, privacy rules such as HIPAA and GDPR make it hard to pool images from several hospitals. Federated Learning (FL) [3] avoids this: the hospitals train a shared model by exchanging model weights, and the images never leave each site.

The main problem in clinical FL is that hospitals see different patients, so their data are not identically distributed (non-IID). Screening clinics see mostly healthy (Grade 0) eyes, referral centres see many severe cases (Grades 3–4), and rural clinics see something in between. Under extreme Dirichlet skew ($\alpha = 0.1$), standard FedAvg collapses: averaged over three seeds, its test QWK drops from 0.647 in its best round to 0.378 in its worst. FedAvg weights each hospital by its number of images, so a few large clinics that each see mostly one grade can take over the global model. FedProx [4] avoids the collapse by limiting how far each local model can move from the global model, but it changes the training at every hospital.

Cross-entropy, the usual training loss, also ignores the order of the grades. For cross-entropy, mistaking Grade 4 for Grade 0 is no worse than mistaking it for Grade 3. Ordinal losses [5]–[6] fix this in centralised training, but little is known about how they behave in FL.

This paper studies **OrdinalFed**, a federated method for ordinal medical grading. It has two parts, and a repeated evaluation tests what each part adds.

1) **Distance-aware OrdinalLoss:** each hospital adds a squared expected-grade penalty to class-weighted cross-entropy, so predictions far from the true grade cost more.
2) **QWK-weighted aggregation with an adaptive $\beta$:** the server scores each client model on a small balanced set $\mathcal{V}$ of 50 images. Over the rounds, it moves the aggregation weights from image counts towards these QWK scores. Nothing changes at the hospitals.
3) **A multi-seed evaluation on held-out test data:** every configuration is run with three seeds and scored on the DDR test split and on the external APTOS 2019 dataset, with ablations and a sensitivity analysis. The results show that OrdinalFed prevents most of FedAvg's collapse. They also show that the gain comes from giving small hospitals more weight, not from the QWK scores or the ordinal loss, and that FedProx is still the most stable method.

## II. Related Work

**Ordinal learning for DR.** Standard multi-class losses treat severity grades as separate categories and ignore their order. Toledo-Cortés et al. [5] proposed Deep Quantum Ordinal Regression (DQOR) and reported a grading mean absolute

error (MAE) of 0.2871 on EyePACS. Polat et al. [6] proposed Class Distance Weighted Cross-Entropy (CDW-CE), which scales the loss by the distance between classes, and reached a QWK of 0.8568 for ulcerative colitis grading. Both studies show that distance-aware losses help in centralised training. As far as the authors know, such losses have not yet been combined with quality-weighted aggregation in federated DR grading.

**Federated learning in ophthalmology.** FedAvg [3] is the standard FL algorithm. FedProx [4] adds an $L_2$ proximal term that keeps local weights close to the global model when client data differ. For DR, Bhulakshmi and Rajput [7] proposed FedDL, which switches between aggregation rules during training, and Ciaramella et al. [8] combined federated DR screening with Grad-CAM explanations across 20 non-IID clients on APTOS 2019. These methods treat grading as plain (nominal) classification, and they weight clients by dataset size rather than by how good their models are.

**Quality-aware aggregation.** Weighting clients by sample count can fail when client data are strongly biased or deliberately corrupted. Bejenar et al. [9] scored client models on a server-side validation set (FedLasso) and beat FedAvg by 45.6 pp under adversarial label flipping. Li et al. [10] showed that client weights that do not sum to one act as an implicit regulariser (FedLAW), and Li et al. [11] proposed FedAAW, which sets aggregation weights from gradient norms and improved non-IID accuracy by 37.32 pp. A review by Mora et al. [12] concluded that client-side regularisation combined with server-side adaptive aggregation gives the most reliable results on non-IID data. OrdinalFed follows this line by pairing an ordinal client loss with QWK-based server aggregation.

## III. Methodology

### A. Problem Formulation

There are $K = 5$ hospital clients. Client $k$ holds a private dataset $\mathcal{D}_k = \{(x_i, y_i)\}_{i=1}^{n_k}$ of fundus images $x_i$ with ordinal grades $y_i \in \{0, 1, 2, 3, 4\}$. A shared EfficientNet-B3 model with weights $w$ is trained over $T$ communication rounds, and no $\mathcal{D}_k$ ever leaves its client. The grade distribution at each client, $P_k(y)$, is very different from the overall distribution $P(y)$, so the local models drift apart during local training.

### B. Limitations of Nominal Objectives

FL clients are usually trained with cross-entropy (CE). With softmax probabilities $\mathbf{p}(x) \in \mathbb{R}^5$, the CE loss for true grade $y$ is $-\log p_y(x)$. This loss treats a prediction of $y + 1$ and a prediction of $y + 4$ the same way: both are just wrong. On a strongly skewed client, such as a clinic that only sees Grade 0 patients, CE soon pushes the local model to predict Grade 0 for every image, and the model forgets what it knew about severe disease.

### C. Distance-Aware OrdinalLoss

To reduce this problem, each client trains with a combined loss, OrdinalLoss:

$$\mathcal{L}_{\text{ord}}(f(x), y) = (1 - \lambda)\, \mathcal{L}_{\text{CE}}(f(x), y) + \lambda\, (\hat{y} - y)^2 \qquad (1)$$

Here $\mathcal{L}_{\text{CE}}$ is cross-entropy with fixed class weights $\mathbf{c} = [1.0, 5.0, 1.5, 10.0, 4.0]$ that offset the class imbalance, and $\lambda = 0.5$ balances the two terms. $\hat{y}$ is the expected grade under the softmax output:

$$\hat{y} = \sum_{c=0}^{4} c \cdot p_c(x) \qquad (2)$$

The penalty $(\hat{y} - y)^2$ grows with the square of the grade error. For a confident prediction, calling a Grade 4 case Grade 0 costs $16\times$ as much as calling it Grade 3. This discourages predictions far from the true grade, even on a client whose data are mostly one grade.

### D. Server Validation Set $\mathcal{V}$ and QWK Estimation

The server cannot see the client data, so it keeps its own small validation set $\mathcal{V}$ of exactly 50 images: 10 per grade, drawn at random (seed 42) from the DDR test split, which shares no images with the clients. Because $\mathcal{V}$ is balanced, every grade counts equally in its score. In round $t$, after it receives the locally updated weights $\tilde{w}_k^t$, the server computes

$$\kappa_k^t = \text{QWK}\big(\{y_i\}_{x_i \in \mathcal{V}},\, \{\hat{y}_k^t(x_i)\}_{x_i \in \mathcal{V}}\big) \qquad (3)$$

### E. Adaptive Cosine Schedule $\beta(t)$

In early rounds, the client models' predictions on $\mathcal{V}$ vary a lot, and each $\kappa_k^t$ is estimated from only 50 images. The aggregation weights therefore move slowly from image counts to quality scores, following a half-cosine schedule:

$$\beta(t) = \beta_0 \cos\left(\frac{\pi(t-1)}{2(T-1)}\right) \qquad (4)$$

where $\beta_0 = 0.5$ and $T = 10$. In the first round, image counts and quality scores count equally ($\beta(1) = 0.5$). In the last round, $\beta(T) = 0$ and only the quality scores are used.

### F. QWK-Weighted Aggregation

The server-set scores are first floored at $\kappa_{\min}$ and turned into quality weights with a temperature-scaled softmax:

$$q_k^t = \frac{\exp\big(\max(\kappa_k^t, \kappa_{\min})/\tau\big)}{\sum_{j=1}^{K} \exp\big(\max(\kappa_j^t, \kappa_{\min})/\tau\big)} \qquad (5)$$

where $\kappa_{\min} = 0.1$ and $\tau = 3$. Each client's aggregation weight mixes this quality weight with the client's share of the images:

$$\omega_k^t = \beta(t)\, \frac{n_k}{\sum_{j=1}^{K} n_j} + \big(1 - \beta(t)\big)\, q_k^t \qquad (6)$$

and the global model is updated as $w^{t+1} = \sum_{k=1}^{K} \omega_k^t \tilde{w}_k^t$. As a safeguard, each client also gets a minimum weight of $\omega_{\min} = 0.03$, followed by renormalisation. With $\beta_0 = 0.5$ and $K = 5$ this never takes effect, because every client already gets at least $(1 - \beta_0)\, q_{\min} \approx 0.08$. Algorithm 1 gives the full procedure.

The scheme is deliberately cautious. No client is ever left out, because one 50-image estimate of $\kappa_k^t$ is too noisy to justify throwing away a hospital's update. With $\tau = 3$ and $\kappa \in [0.1, 1]$, the quality weights $q_k^t$ differ from each other by at most a factor of $e^{0.9/3} \approx 1.35$, so they are close to

equal. As $\beta(t)$ falls, the aggregation therefore moves from image-count weights towards almost equal client weights, with a slight preference for clients that score higher on $\mathcal{V}$. Small clients gain influence in later rounds. The uniform-aggregation ablation ($\omega_k^t = 1/K$, Sec. IV-D) separates this effect from the effect of the QWK scores.

## IV. Experimental Setup

### A. Dataset and Preprocessing

All methods are evaluated on the **DDR dataset** [2], which contains 13,673 fundus photographs, of which 12,522 are gradable. After removing the ungradable images, 6,260 training, 2,503 validation and 3,759 test images remain. All images are resized to $224 \times 224$ pixels. During local training, random horizontal flips, rotations of up to $\pm 15°$, and mild colour jitter are applied to reduce overfitting. The training set is strongly imbalanced: Grade 0 (50.0%), Grade 1 (5.0%), Grade 2 (35.8%), Grade 3 (1.9%), and Grade 4 (7.3%). For external validation, the APTOS 2019 dataset [13] is used: 3,662 labelled fundus photographs collected in India and graded on the same 0–4 scale. A public copy resized to $224 \times 224$ pixels is used, and these images are never used for training or model selection.

### B. Modeling Clinical Heterogeneity

The data are split across $K = 5$ clients in three ways, shown in Fig. 1. In the figure, each bar is one client. Its segments show the share of each DR grade among that client's images, from light (Grade 0) to dark (Grade 4), and the number under each client label is the client's total number of images.

- **IID partition:** images are assigned uniformly at random (1,252 per client). This is the easiest case for FL.
- **Dirichlet non-IID ($\alpha = 0.1$):** the skew is generated with a Dirichlet distribution $p_k \sim \mathrm{Dir}(\alpha)$. With $\alpha = 0.1$, most clients draw their images almost entirely from a single grade. Client sizes range from 32 to 2,087 images: Client 0 holds only 32 images (100% Grade 0), and Client 4 holds 2,087 images, 98% of them Grade 2.
- **Real-site clinical partition:** images are assigned using fixed grade profiles that model five types of clinic: Site 0 (general screening clinic; 40% of images, 84% Grade 0), Site 1 (rural primary care; 22%, mainly Grades 0 and 2, with the largest Grade 1 share at 11%), Site 2 (specialist referral eye hospital; 10%, the largest Grade 3–4 share at 38%), Site 3 (diabetic retinopathy clinic; 19%, 67% Grade 2), and Site 4 (urban hospital; 9%, mixed grades).

### C. Network Architecture and Baselines

All federated experiments use an EfficientNet-B3 backbone, which offers a good balance between model size and accuracy at $224 \times 224$ resolution. As in a real deployment that adapts pre-trained weights, all FL runs start from a centralised model trained for 30 epochs (best validation QWK = 0.834, reached at epoch 23). OrdinalFed is compared with: (1) a **Centralised Baseline**; (2) **FedAvg** [3]; (3) **FedProx** [4] with proximal coefficient $\mu = 0.01$; and (4) **AccWt**, an accuracy-weighted baseline based on server-side validation scoring [9]. AccWt uses the same schedule and softmax weighting as OrdinalFed (Eqs. 4–6), but it scores each client by plain accuracy on $\mathcal{V}$ instead of QWK, and its clients train with class-weighted cross-entropy. A fifth configuration, **OrdinalFed + Prox**, adds FedProx's proximal term ($\mu = 0.01$) to the OrdinalFed clients. It tests whether the server-side and client-side fixes work well together. All methods run $T = 10$ communication rounds with $E = 5$ local epochs, batch size 64, the Adam optimiser ($\eta = 10^{-4}$), and the same class weights $\mathbf{c}$.

**Algorithm 1** OrdinalFed Training Pipeline

**Require:** Clients $K$, rounds $T$, local epochs $E$, trade-off $\lambda$, initial schedule $\beta_0$, server set $\mathcal{V}$, initial weights $w^0$.

**for** $t = 1$ **to** $T$ **do**
  Broadcast global model $w^t$ to all clients $k \in \{1, \ldots, K\}$
  **for** each client $k \in \{1, \ldots, K\}$ **in parallel do**
    $\tilde{w}_k^t \leftarrow \textsc{LocalTrain}(w^t, \mathcal{D}_k, E, \lambda)$ *(Eq. 1)*
    Evaluate validation score $\kappa_k^t$ on $\mathcal{V}$ *(Eq. 3)*
  **end for**
  Compute annealing factor $\beta(t)$ *(Eq. 4)*
  Compute aggregation weights $\omega_k^t$ for all $k$ *(Eqs. 5–6)*
  $w^{t+1} \leftarrow \sum_{k=1}^{K} \omega_k^t \tilde{w}_k^t$
**end for**
**return** Final model $w^T$

### D. Ablations and Evaluation Protocol

Three ablations show what each part of OrdinalFed contributes: (i) OrdinalLoss with sample-count aggregation; (ii) class-weighted cross-entropy with QWK-weighted aggregation; and (iii) class-weighted cross-entropy with uniform aggregation ($\omega_k^t = 1/K$). Three settings of OrdinalFed are also tested under $\alpha = 0.1$. Each changes one value of the default ($\lambda = 0.5$, $\beta_0 = 0.5$, $\tau = 3$): a weaker ordinal term ($\lambda = 0.1$), quality weights from the first round ($\beta_0 = 0$), and sharper quality weights ($\tau = 0.5$). These settings were fixed before the runs and are compared on mean validation QWK only.

Every configuration is repeated with three random seeds, which control data order and augmentation. Results are reported as mean $\pm$ standard deviation. GPU arithmetic is not bit-exact, so runs with the same seed can differ slightly. After each round, the global model is evaluated on the DDR validation split and on the DDR test split without the 50 images of $\mathcal{V}$ (3,709 images). For each run, the reported model is the one from the round with the highest validation QWK. The test split is never used for training, aggregation or model selection, so it gives an independent estimate of performance. The same models are also scored on APTOS 2019 without retraining. Stability is summarised by the mean, worst and final test QWK over the 10 rounds.

## V. Results and Discussion

### A. Overall Performance

Table I gives the test QWK of each method at its best validation round, averaged over three seeds. The federated methods score between 0.640 and 0.681, close to the 0.682 of

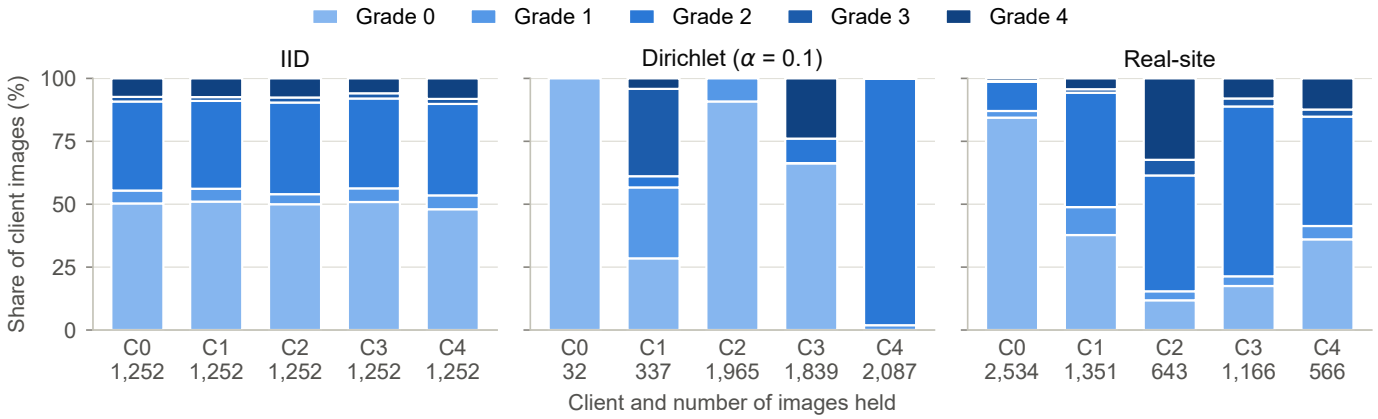



**Fig. 1:** Grade mix of each client under the IID, Dirichlet ($\alpha = 0.1$) and real-site partitions. Under $\alpha = 0.1$, client sizes range from 32 images (Client 0, 100% Grade 0) to 2,087 images (Client 4, 98% Grade 2). The real-site partition models five types of clinic, from screening clinics to specialist referral centres.

the centralised model. On the IID and real-site partitions, the methods differ by at most 0.01, which is within the variation between seeds. Under $\alpha = 0.1$, FedProx scores highest (0.664) and OrdinalFed lowest (0.640). This table hides what happens in the other rounds, because the reported round is chosen on the validation split. For FedAvg under $\alpha = 0.1$, for example, the best validation round is always the first one, before the collapse.

### *B. Stability Under Extreme Skew*

Fig. 2 shows the test QWK of the global model after every round, with one panel per partition. Lines show the mean over three seeds, shaded bands ± one standard deviation, and the dashed line the test QWK of the centralised model. On the IID and real-site partitions, all three methods stay close to the centralised line. Under $\alpha = 0.1$, FedAvg collapses from round 5 onwards, FedProx stays flat, and OrdinalFed avoids the collapse but goes up and down from round to round.

Table II summarises the ten rounds under $\alpha = 0.1$. Compared with FedAvg, OrdinalFed raises the worst round from 0.378 to 0.514 and the mean over rounds from 0.535 to 0.604, with no change at the hospitals. FedProx, which changes local training instead, is the most stable method (worst round 0.641). Adding FedProx's proximal term to OrdinalFed brings it close to FedProx (worst round 0.627) and gives the best final round (0.664), but it is not more stable than FedProx alone.

### *C. What Makes the Aggregation Stable?*

Table III separates the two parts. OrdinalLoss alone does not stop the collapse: with image-count weights, its worst round (0.380) is the same as FedAvg's. QWK-weighted aggregation with cross-entropy raises the worst round to 0.586, but giving every client the same weight raises it further, to 0.630. Scoring clients by accuracy instead of QWK (AccWt, Table II) gives almost the same result as scoring them by QWK (worst round 0.571 versus 0.586). On the real-site partition, where no method collapses, the four ablation configurations stay within

**TABLE I:** Test QWK at the Best Validation Round (Mean ± Std over Three Seeds). Bold: Best Federated Method per Column.

| Method | IID | Dirichlet ($\alpha = 0.1$) | Real-Site |
|---|---|---|---|
| Centralised (reference) | 0.682 | 0.682 | 0.682 |
| FedAvg [3] | 0.677 ±.008 | 0.647 ±.010 | 0.673 ±.004 |
| FedProx [4] | **0.681** ±.014 | **0.664** ±.015 | 0.671 ±.005 |
| AccWt | – | 0.651 ±.011 | **0.674** ±.009 |
| OrdinalFed | 0.679 ±.006 | 0.640 ±.007 | 0.666 ±.002 |
| OrdinalFed + Prox | – | 0.659 ±.012 | 0.672 ±.007 |

**TABLE II:** Test QWK Across the 10 Rounds Under Dirichlet Skew ($\alpha = 0.1$): Mean over Rounds, Worst Round and Final Round. Bold: Best per Column.

| Method | Mean | Worst | Final |
|---|---|---|---|
| FedAvg | 0.535 ±.004 | 0.378 ±.035 | 0.475 ±.026 |
| FedProx | **0.659** ±.003 | **0.641** ±.008 | 0.650 ±.006 |
| AccWt | 0.627 ±.011 | 0.571 ±.017 | 0.657 ±.018 |
| OrdinalFed | 0.604 ±.005 | 0.514 ±.027 | 0.652 ±.016 |
| OrdinalFed + Prox | 0.651 ±.003 | 0.627 ±.005 | **0.664** ±.013 |

0.02 of each other (test QWK at the best validation round between 0.663 and 0.683).

The logged aggregation weights explain this. Under $\alpha = 0.1$, the two smallest clients hold 5.9% of the images, so FedAvg gives them 5.9% of the weight. The quality weights of Eq. (5) are close to equal, so OrdinalFed raises their combined weight from 23.5% in round 1 to 40.8% in round 10. The QWK scores themselves carry little information at this scale. A client's score on the 50-image set $\mathcal{V}$ changes by 0.13 on average from one round to the next, and in round 10 the largest quality weight is only 1.19 times the smallest. The stability therefore comes from moving weight towards small clients, not from the QWK scores. OrdinalLoss lowers performance slightly in both comparisons.

The lower block of Table III tests three settings of Ordi-

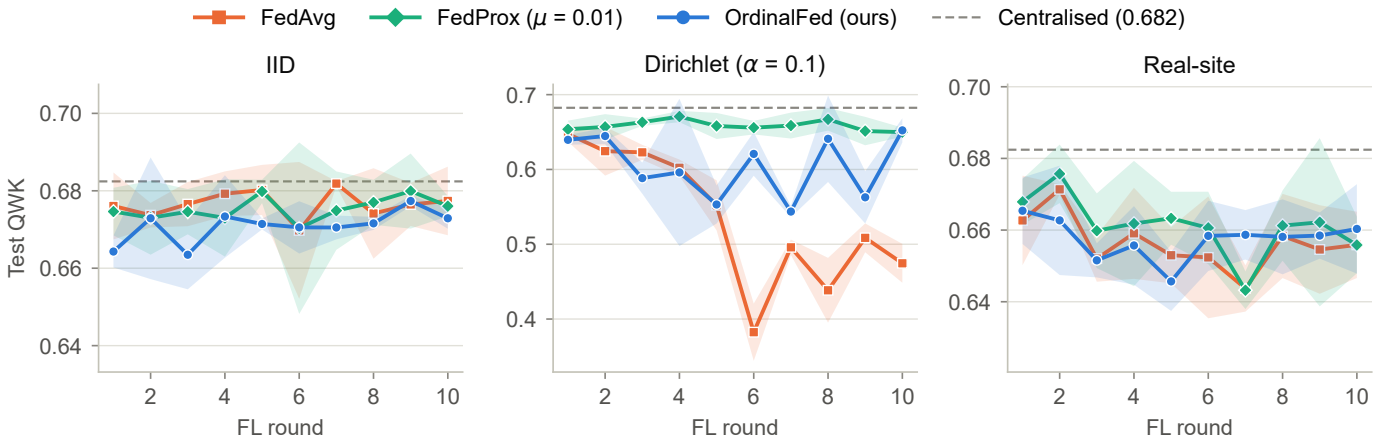



**Fig. 2:** Test QWK of the global model after each of the 10 rounds for the IID, Dirichlet ($\alpha = 0.1$) and real-site partitions (mean $\pm$ std over three seeds). Under $\alpha = 0.1$, FedAvg collapses, OrdinalFed avoids the collapse but fluctuates, and FedProx stays stable.

nalFed, each fixed before the runs and compared on mean validation QWK. All three do better than the default. Sharper quality weights ($\tau = 0.5$) score best (0.795 versus 0.761), closely followed by using quality weights from the first round ($\beta_0 = 0$, 0.793). Both still score below equal weighting (0.810) and FedProx (0.808).

### *D. Class-Wise Performance*

Table IV gives the per-grade recall and F1 on the test split. All methods find Grade 0 reliably (recall $\geq 0.95$) but almost never find Grade 1 (recall $\leq 0.05$), which makes up only 5% of the training set. Grade 2 recall stays near 0.4 because many test Grade 2 images are predicted as Grade 0 (Sec. V-G). OrdinalFed has the highest Grade 3 recall in both partitions (0.36 versus 0.27–0.31 under $\alpha = 0.1$, and 0.40 versus 0.35–0.38 on the real-site partition). This is a modest gain on the rarest severe grade. For Grade 4, the three methods are within 0.05 of each other.

### *E. External Validation*

To test the models on data from a different source, the best-validation model of every run is also scored on APTOS 2019 [13], without retraining (Table V). The centralised model reaches a QWK of 0.641 there, against 0.682 on the DDR test split. The federated models score between 0.597 and 0.658, so they transfer about as well as the centralised model. Under $\alpha = 0.1$, OrdinalFed has the highest mean (0.651) and the smallest spread across seeds of the methods trained on that partition. The differences between methods, however, are within the seed-to-seed variation (standard deviations of up to 0.057). Because each run is scored at its best validation round, these numbers do not show FedAvg's later collapse.

### *F. Communication and Computational Overhead*

OrdinalFed sends the same model weights as FedAvg, so it adds no communication. Once per round, the server evaluates each of the $K$ client models on the 50 images of $\mathcal{V}$. This took a median of 1.2 s per round for $K = 5$ on an Intel Arc B580 GPU, about 0.6% of the 195 s a full round takes.

**TABLE III:** Contribution of Each Component and Setting Under $\alpha = 0.1$. Mean: QWK Averaged over the 10 Rounds; Worst: Lowest Round. CE: Class-Weighted Cross-Entropy. Default: $\lambda = 0.5$, $\beta_0 = 0.5$, $\tau = 3$.

| Configuration | Mean val | Mean test | Worst test |
|---|---|---|---|
| CE + image-count weights (FedAvg) | 0.670 $\pm$.011 | 0.535 $\pm$.004 | 0.378 $\pm$.035 |
| OrdinalLoss + image-count weights | 0.659 $\pm$.015 | 0.522 $\pm$.011 | 0.380 $\pm$.018 |
| CE + QWK weights | 0.786 $\pm$.006 | 0.633 $\pm$.010 | 0.586 $\pm$.006 |
| CE + equal weights | **0.810** $\pm$.007 | **0.658** $\pm$.006 | **0.630** $\pm$.021 |
| OrdinalFed (default) | 0.761 $\pm$.003 | 0.604 $\pm$.005 | 0.514 $\pm$.027 |
| with $\lambda = 0.1$ | 0.776 $\pm$.007 | 0.622 $\pm$.006 | 0.571 $\pm$.019 |
| with $\beta_0 = 0$ | 0.793 $\pm$.005 | 0.631 $\pm$.009 | 0.571 $\pm$.011 |
| with $\tau = 0.5$ | 0.795 $\pm$.006 | 0.637 $\pm$.005 | 0.592 $\pm$.009 |

**TABLE IV:** Per-Grade Recall / F1 on the Test Split at the Best Validation Round (Mean over Three Seeds).

| Method | G0 | G1 | G2 | G3 | G4 |
|---|---|---|---|---|---|
| *Dirichlet ($\alpha = 0.1$)* | | | | | |
| FedAvg | .96/.80 | .05/.07 | .41/.52 | .27/.32 | .66/.73 |
| FedProx | .95/.81 | .05/.06 | .43/.54 | .31/.34 | .69/.74 |
| OrdinalFed | .97/.80 | .04/.07 | .39/.50 | .36/.38 | .64/.73 |
| *Real-site* | | | | | |
| FedAvg | .98/.81 | .04/.06 | .44/.55 | .35/.37 | .66/.73 |
| FedProx | .98/.81 | .04/.06 | .43/.54 | .38/.38 | .65/.73 |
| OrdinalFed | .97/.81 | .04/.06 | .43/.54 | .40/.38 | .64/.73 |

### *G. Limitations*

Grade 1 recall stays at or below 0.05 for every method (Table IV). All models also score lower on the DDR test split than on the validation split. The centralised model drops from a QWK of 0.834 to 0.682, and the federated models drop by a similar amount. Most of this drop comes from Grade 2 images predicted as Grade 0 (about 45% of test Grade 2 images, against about 12% on the validation split), and the test split

**TABLE V:** External Validation on APTOS 2019 (3,662 Images, Never Used for Training or Selection): QWK of Each Run's Best-Validation Model. Columns Give the Training Partition. Bold: Best Federated Method per Column.

| **Method** | **IID** | **Dirichlet** ($\alpha = 0.1$) | **Real-Site** |
|---|---|---|---|
| Centralised (reference) | 0.641 | 0.641 | 0.641 |
| FedAvg | **0.615** $\pm$.030 | 0.623 $\pm$.057 | 0.656 $\pm$.013 |
| FedProx | 0.600 $\pm$.006 | 0.607 $\pm$.043 | **0.658** $\pm$.018 |
| AccWt | – | 0.645 $\pm$.038 | 0.637 $\pm$.014 |
| OrdinalFed | 0.599 $\pm$.022 | **0.651** $\pm$.018 | 0.641 $\pm$.056 |
| OrdinalFed + Prox | – | 0.628 $\pm$.048 | 0.597 $\pm$.013 |

has a different mix of camera resolutions. Validation QWK therefore overestimates performance on new images, which is why all main results are reported on the test split. The 50-image server set $\mathcal{V}$ is too small to rank clients reliably, which limits the value of QWK-based weighting, and it needs labelled images on the server. Finally, all models were trained on one dataset at $224 \times 224$ resolution with five clients, and APTOS 2019 is the only external test set. Larger server sets, higher resolution, more clients and more external datasets are left for future work.

## VI. Conclusion

This paper evaluated OrdinalFed, a federated method for multi-grade Diabetic Retinopathy screening that combines a distance-aware OrdinalLoss with QWK-weighted aggregation. All experiments used the DDR dataset, three seeds and a held-out test split, and the trained models were also tested on the external APTOS 2019 dataset, where they scored close to the centralised model. Under extreme label skew, OrdinalFed's server-side weighting prevents most of FedAvg's collapse (worst-round test QWK 0.514 versus 0.378) without extra communication. The ablations show why: small hospitals get more weight. Equal client weights work even better, a 50-image server set is too small for the QWK scores to rank clients, and the OrdinalLoss does not help in the federated setting. FedProx remains the most stable method, and adding its proximal term to OrdinalFed comes close to FedProx but does not beat it. On realistic multi-site and IID partitions, all methods perform about the same. Future work will test larger server sets, higher image resolution and more external datasets, and will look at formal differential privacy guarantees.

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